Panagiotis Toloudis

Integrated Master's Degree
Electrical & Computer Engineering
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Summary

High-performing **Integrated Master's ECE student** (GPA: 7.67/10) specializing in **Distributed Systems, MLOps, and Cloud Infrastructure**. Proven expertise in low-level system programming (Linux Kernel) and scalable multi-node environments. Currently spearheading IT infrastructure optimizations for the Greek Army's Research & IT office, managing critical systems for 50+ users.

Experience

- **Research & IT Office, Greek Army** — *Sep. 2025 – Present*
  IT Specialist / System Administrator — Greece

  - Administer Windows Active Directory (AD) for 50+ users, implementing Group Policy Objects (GPOs) to automate security configurations.
  - Optimize hardware reliability by executing diagnostics and repairs on critical servers and network peripherals.

Education

- **University of Thessaly** — *2019 – 2025*
  Integrated Master in Electrical & Computer Engineering (GPA: 7.67/10) — Volos, Greece

Positions of Responsibility

- **University of Thessaly** — *Oct. 2023 – Jan. 2024*
  Graduate Teaching Assistant — Volos, Greece

  - **Machine Learning for Data Science (10/10 Grade):** Evaluated large-scale student implementations of neural networks and regression models, ensuring 100% adherence to MLOps best practices.
  - **Object-Oriented Programming (9.5/10 Grade):** Directed technical laboratories for 50+ students and architected automated Java grading scripts, resulting in a 15% increase in average assignment scores.

Key Certifications

- **Cloud & Infrastructure**: Kubernetes (LFS158), AWS Cloud 101, Azure Kubernetes Service (AKS), Docker Containerization, Kubeflow (LFS147), DevOps & SRE (LFS162).
- **Security & Systems**: Zero Trust Security Strategy (Microsoft & LFS183), Linux Kernel Development (LFD103), Embedded Systems (LFD116), Windows PowerShell.
- **AI & Programming**: PyTorch & Deep Learning (LFS116), Rust Language Development, MLOps, Unity Junior Programmer.

Technical Skills

- **Data Engineering & Databases**: Advanced SQL (PostgreSQL, MySQL), NoSQL (MongoDB), ETL Pipelines, **Database Management Systems (Optimization & Indexing)**, Data Mining.
- **Cloud, MLOps & DevOps**: Kubernetes (Certified LFS158), Docker, Kubeflow, Azure (AKS), AWS, CI/CD, Virtualization.
- **AI & Programming**: Python (Pandas, NumPy, Scikit-learn), PyTorch, TensorFlow, C/C++, Rust, Go, Java, Bash.
- **Distributed & Parallel Systems**: Distributed Execution Environments, **OpenMP**, Multi-threading, Load Balancing.
- **Infrastructure & Security**: Linux Kernel Development, **Active Directory (AD)**, Windows Server, Zero Trust Networking, Hardware Diagnostics.
- **Developer Tools**: Git/GitHub, GDB, Valgrind, Unity, Unreal Engine, LaTeX.

Selected Technical Projects

•Simulated Quadcopter Drone (Diploma Thesis) *2024*

C#, Python, Ardupilot: Developed a real-time bridge using Unity as an external Flight Dynamic Model (FDM) for 1:1 accuracy. GitHub

•Linux Kernel & Memory Management *Apr. 2023*

C, Linux Kernel: Optimized SLOB allocator for First-Fit strategy and integrated custom System Calls into the kernel. GitHub

•Parallel Fast Fourier Transform (FFT) *Fall 2022*

C, OpenMP: Engineered the FFT algorithm utilizing multi-threading and SIMD vectorization, achieving a significant computational speedup on multi-core Linux systems through optimized memory access patterns and thread load balancing. GitHub

•Distributed Execution Environment *Spring 2022*

Python: Architected a multi-node system with transparent process migration, increasing task throughput by 25%. GitHub

Volunteering & Leadership

- **AgroJason Project | Project Manager:** Led a cross-functional team of 10+ students to design an IoT greenhouse automation system using Arduino and Raspberry Pi.

- **IEEE Student Branch | Board Member:** Co-organized 3+ technical workshops and contributed to ECESCON 2023 and SMACD 2024 international conferences.

Languages

- **English**: TOEIC B2 Proficiency.
- **German**: B2 Level (In Progress) - Professional working proficiency.
- **Greek**: Native Speaker.